MATH 526 Section 1 Lab 9 Spring 2006

Part I

In this part we solve a 3×3 system by the Jacobi method.

The Jacobi method for solving $\mathbf{Ax} = \mathbf{b}$ is defined by the following iterative formula:

$$\mathbf{S}\mathbf{x}^{\text{new}} = \mathbf{b} + \mathbf{T}\mathbf{x}^{\text{old}}$$

where $\mathbf{S}$ is the diagonal part of $\mathbf{A}$ and $\mathbf{T} = \mathbf{S} - \mathbf{A}$.

Create

$$\mathbf{A} = \begin{bmatrix} 5 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 5 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 7 \\ 7 \\ 7 \end{bmatrix} .$$

The exact solution for $\mathbf{Ax} = \mathbf{b}$ is clearly

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} .$$

The following commands will compute 10 Jacobi iterations with initial guess $\mathbf{0}$.

```
>>x= zeros(3,1)
>>S = diag(diag(A))
>>T = S - A
>>for k=1:10
>>x = S\(b+T*x)
>>end
```

You can see that the sequence of vectors generated by Jacobi's method is converging to the exact solution.

```
>>clear x
```

You can also iterate until an approximate solution that satisfies a preset tolerance for the estimated relative error is obtained, or a preset upper bound for the number of iterations is reached.

Create the following script M-file in an M-file window and store it as jacobi1.m (the comments are optional).

```
tol=1.e-6;                          % Set the tolerance.
y=zeros(3,1);                       % Set the initial guess.
x=S\(b+T*y);                        % Compute the first iteration.
niter=1;                            % Set the count for the number of iterations.
re=norm(x-y,inf)/norm(x,inf);       % Compute the first estimate for the relative
                                    % error in the ∞-norm.
while re>tol & niter<100
   y=x;
   x=S\(b+T*y);
   re=norm(x-y,inf)/norm(x,inf);
   niter=niter+1;
end
```

Go back to the MATLAB command window and type

\>>**format long e**
\>>**jacobi1**

The lines in jacobi1.m have been executed. To see the results, type the following commands.

\>>**x**
\>>**re**
\>>**niter**

Clear all variables before you work on Part II.